

ANDROID STUDIO

1. FontColorToastApp –

Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/textView"
        android:text="Hello Android!"
        android:textSize="24sp"
        android:textStyle="normal"
        android:textColor="#000000"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/button"
        android:text="Change Style"
        android:layout_marginTop="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

MainActivity.java:

```
package com.example.fontcolortoastapp;
import android.graphics.Color;
import android.graphics.Typeface;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
```

```

TextView textView;
Button button;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    textView = findViewById(R.id.textView);
    button = findViewById(R.id.button);

    button.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {

            // Change text color
            textView.setTextColor(Color.RED);

            // Change font style
            textView.setTypeface(null, Typeface.BOLD_ITALIC);

            // Show Toast message
            Toast.makeText(MainActivity.this,
                "Font and Color Changed!",
                Toast.LENGTH_SHORT).show();
        }
    });
}

```

AndroidManifest.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportRtl="true"

```

```

        android:theme="@style/Theme.FontColorToastApp">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>

```

2. CalculatorApp –

Activity_main.xml:

```

    <?xml version="1.0" encoding="utf-8"?>
    <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
        android:orientation="vertical"
        android:padding="20dp"
        android:layout_width="match_parent"
        android:layout_height="match_parent">

        <EditText
            android:id="@+id/num1"
            android:hint="Enter First Number"
            android:inputType="numberDecimal"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <EditText
            android:id="@+id/num2"
            android:hint="Enter Second Number"
            android:inputType="numberDecimal"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginTop="10dp"/>

        <TextView
            android:id="@+id/result"
            android:text="Result"
            android:textSize="20sp"

```

```
        android:layout_marginTop="20dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
```

```
<Button
    android:text="Add"
    android:onClick="add"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
<Button
    android:text="Subtract"
    android:onClick="subtract"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
<Button
    android:text="Multiply"
    android:onClick="multiply"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
<Button
    android:text="Divide"
    android:onClick="divide"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
</LinearLayout>
```

MainActivity.java:

```
package com.example.calculatorapp;

import android.os.Bundle;
import android.view.View;
import android.widget.EditText;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {

    EditText num1, num2;

    TextView result;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        num1 = findViewById(R.id.num1);
        num2 = findViewById(R.id.num2);
        result = findViewById(R.id.result);
    }

    public void add(View v) {
        double a = Double.parseDouble(num1.getText().toString());
        double b = Double.parseDouble(num2.getText().toString());
        result.setText("Result: " + (a + b));
    }

    public void subtract(View v) {
        double a = Double.parseDouble(num1.getText().toString());
        double b = Double.parseDouble(num2.getText().toString());
        result.setText("Result: " + (a - b));
    }

    public void multiply(View v) {
        double a = Double.parseDouble(num1.getText().toString());
```

```

        double b = Double.parseDouble(num2.getText().toString());
        result.setText("Result: " + (a * b));
    }

    public void divide(View v) {
        double a = Double.parseDouble(num1.getText().toString());
        double b = Double.parseDouble(num2.getText().toString());

        if (b == 0) {
            result.setText("Cannot divide by zero");
        } else {
            result.setText("Result: " + (a / b));
        }
    }
}

```

3. BankApp –

Activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="24dp"
    android:background="#F4F6F9">

    <TextView
        android:text="BANKAPP LOGIN"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_marginBottom="30dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

```

```
<EditText
    android:id="@+id/emailInput"
    android:hint="Email"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
<EditText
    android:id="@+id/passwordInput"
    android:hint="Password"
    android:inputType="textPassword"
    android:layout_marginTop="15dp"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
<Button
    android:id="@+id/loginBtn"
    android:text="Login"
    android:layout_marginTop="20dp"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
<Button
    android:id="@+id/registerBtn"
    android:text="Register New Account"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:backgroundTint="#2a5298"/>
```

```
</LinearLayout>
```

Activity_dashboard.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#f5f5f5">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">
```

```
<!-- Header -->
<TextView

    android:layout_width="wrap_content"
    android:layout_height="wrap_content"

    android:textSize="24sp"
    android:textStyle="bold"
    android:layout_gravity="center"/>

<!-- Current Balance -->
<TextView
    android:id="@+id/balanceText"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Current Balance: ₹0"
    android:textSize="18sp"
    android:textStyle="bold"
    android:layout_marginTop="12dp"
    android:layout_gravity="center"/>

<!-- Amount Input -->
<EditText
    android:id="@+id/amountInput"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Amount"
    android:inputType="numberDecimal"
    android:layout_marginTop="20dp"/>

<Button
    android:id="@+id/depositBtn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Deposit"
    android:layout_marginTop="10dp"/>

<Button
    android:id="@+id/withdrawBtn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Withdraw"
    android:layout_marginTop="10dp"/>
```

```
<Button
    android:id="@+id/transactionsToggleBtn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Transactions"
    android:layout_marginTop="20dp"/>
```

```
<ListView
    android:id="@+id/listView"
    android:layout_width="match_parent"
    android:layout_height="250dp"
    android:visibility="gone"
    android:layout_marginTop="10dp"/>
```

```
<Button
    android:id="@+id/downloadPdfBtn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Download Statement (PDF)"
    android:layout_marginTop="20dp"/>
```

```
<Button
    android:id="@+id/logoutBtn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Logout"
    android:layout_marginTop="10dp"/>
```

```
</LinearLayout>
```

```
</ScrollView>
```

MainActivity.java:

```
package com.example.bankapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.*;
import android.content.Intent;
import android.view.View;
import java.io.*;
import java.net.*;
```

```

import android.util.Log;

public class MainActivity extends AppCompatActivity {

    EditText emailInput, passwordInput;
    Button loginBtn, registerBtn; // ← Added registerBtn here
    TextView registerLink;

    // IMPORTANT: CHANGE THIS TO YOUR COMPUTER'S IP ADDRESS
    // For emulator: "http://10.0.2.2:8080/BankWebApp/"
    // For real phone: "http://192.168.1.5:8080/BankWebApp/" (use your actual IP)
    private final String BASE_URL = "http://10.187.85.203:8080/BankWebApp/";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        emailInput = findViewById(R.id.emailInput);
        passwordInput = findViewById(R.id.passwordInput);
        loginBtn = findViewById(R.id.loginBtn);
        registerBtn = findViewById(R.id.registerBtn); // ← Added this line

        loginBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                loginUser();
            }
        });

        // ← ADD THE REGISTER BUTTON CODE HERE
        registerBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // Open registration in browser
                Intent intent = new Intent(Intent.ACTION_VIEW);
                intent.setData(android.net.Uri.parse(BASE_URL + "bank.html"));
                startActivity(intent);
            }
        });
    }

    private void loginUser() {

```

```

final String email = emailInput.getText().toString().trim();
final String password = passwordInput.getText().toString().trim();

if (email.isEmpty() || password.isEmpty()) {
    Toast.makeText(this, "Enter Email and Password",
Toast.LENGTH_SHORT).show();
    return;
}

// Network operation must be in background thread
new Thread(new Runnable() {
    @Override
    public void run() {
        try {
            URL url = new URL(BASE_URL + "login");
            HttpURLConnection conn = (HttpURLConnection) url.openConnection();
            conn.setRequestMethod("POST");
            conn.setDoOutput(true);
            conn.setRequestProperty("User-Agent", "Android");
            conn.setRequestProperty("Content-Type", "application/x-www-form-
urlencoded");

            // Send login data
            String postData = "email=" + email + "&password=" + password;
            OutputStream os = conn.getOutputStream();
            os.write(postData.getBytes());
            os.flush();
            os.close();

            // Check response code
            int responseCode = conn.getResponseCode();

            // Get session cookie (if any)
            String cookie = conn.getHeaderField("Set-Cookie");
            if (cookie != null) {
                // Store just the JSESSIONID part
                String[] cookieParts = cookie.split(";");
                NetworkUtils.SESSION_COOKIE = cookieParts[0];
                Log.d("SESSION", "Cookie stored: " + NetworkUtils.SESSION_COOKIE);
            }

            // Read response
            BufferedReader br;
            if (responseCode == 200) {

```

```

        br = new BufferedReader(new
InputStreamReader(conn.getInputStream()));
    } else {
        br = new BufferedReader(new
InputStreamReader(conn.getErrorStream()));
    }

    final String response = br.readLine();
    br.close();
    conn.disconnect();

    Log.d("LOGIN_RESPONSE", response);

    // Update UI on main thread
    runOnUiThread(new Runnable() {
        @Override
        public void run() {
            if (response != null && response.contains("SUCCESS")) {
                Toast.makeText(MainActivity.this, "Login Successful",
Toast.LENGTH_SHORT).show();
                // Go to Dashboard
                Intent intent = new Intent(MainActivity.this,
DashboardActivity.class);
                startActivity(intent);
                finish();
            } else {
                String errorMsg = response != null ? response.replace("ERROR:", "")
: "Login Failed";
                Toast.makeText(MainActivity.this, errorMsg,
Toast.LENGTH_LONG).show();
            }
        }
    });

    } catch (final Exception e) {
        e.printStackTrace();
        runOnUiThread(new Runnable() {
            @Override
            public void run() {
                Toast.makeText(MainActivity.this, "Connection Error: " +
e.getMessage(), Toast.LENGTH_LONG).show();
            }
        });
    }
}

```

```

    }
    }).start();
}
}

```

DashboardActivity.java:

```

package com.example.bankapp;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;
import org.json.*;

import java.io.*;
import java.net.*;

public class DashboardActivity extends AppCompatActivity {

    TextView balanceText;
    EditText amountInput;
    Button depositBtn, withdrawBtn, logoutBtn,
        transactionsToggleBtn, downloadPdfBtn;
    ListView listView;

    ArrayAdapter<String> adapter;
    boolean transactionsVisible = false;

    private final String BASE_URL =
        "http://10.0.2.2:8080/BankWebApp/";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_dashboard);

        balanceText = findViewById(R.id.balanceText);
        amountInput = findViewById(R.id.amountInput);
        depositBtn = findViewById(R.id.depositBtn);
        withdrawBtn = findViewById(R.id.withdrawBtn);
        logoutBtn = findViewById(R.id.logoutBtn);
        transactionsToggleBtn = findViewById(R.id.transactionsToggleBtn);
    }
}

```

```

downloadPdfBtn = findViewById(R.id.downloadPdfBtn);
listView = findViewById(R.id.listView);

adapter = new ArrayAdapter<>(this,
    android.R.layout.simple_list_item_1);
listView.setAdapter(adapter);

loadBalance();

depositBtn.setOnClickListener(v -> transact("deposit"));
withdrawBtn.setOnClickListener(v -> transact("withdraw"));

transactionsToggleBtn.setOnClickListener(v -> {
    if (transactionsVisible) {
        listView.setVisibility(ListView.GONE);
        transactionsVisible = false;
    } else {
        listView.setVisibility(ListView.VISIBLE);
        loadTransactions();
        transactionsVisible = true;
    }
});

downloadPdfBtn.setOnClickListener(v ->
    startActivity(new Intent(Intent.ACTION_VIEW,
        Uri.parse(BASE_URL + "downloadStatement"))));

logoutBtn.setOnClickListener(v -> {
    NetworkUtils.SESSION_COOKIE = null;
    startActivity(new Intent(this, MainActivity.class));
    finish();
});
}

private void transact(String type) {

    String amount = amountInput.getText().toString().trim();

    if (amount.isEmpty()) {
        Toast.makeText(this, "Enter amount", Toast.LENGTH_SHORT).show();
        return;
    }

    new Thread(() -> {

```

```

try {

    URL url = new URL(BASE_URL + type);
    HttpURLConnection conn = (HttpURLConnection) url.openConnection();
    conn.setRequestMethod("POST");
    conn.setDoOutput(true);

    if (NetworkUtils.SESSION_COOKIE != null) {
        conn.setRequestProperty("Cookie",
            NetworkUtils.SESSION_COOKIE);
    }

    OutputStream os = conn.getOutputStream();
    os.write(("amount=" + amount).getBytes());
    os.flush();
    os.close();

    BufferedReader br =
        new BufferedReader(new InputStreamReader(conn.getInputStream()));
    String response = br.readLine();
    br.close();

    runOnUiThread(() -> {
        Toast.makeText(this, response, Toast.LENGTH_SHORT).show();
        amountInput.setText("");
        loadBalance();
        if (transactionsVisible) loadTransactions();
    });

} catch (Exception e) {
    e.printStackTrace();
}
}).start();
}

private void loadBalance() {

    new Thread(() -> {
        try {

            URL url = new URL(BASE_URL + "balance");
            HttpURLConnection conn = (HttpURLConnection) url.openConnection();

            if (NetworkUtils.SESSION_COOKIE != null) {

```

```

        conn.setRequestProperty("Cookie",
                                NetworkUtils.SESSION_COOKIE);
    }

    BufferedReader br =
        new BufferedReader(new InputStreamReader(conn.getInputStream()));
    String balance = br.readLine();
    br.close();

    runOnUiThread(() ->
        balanceText.setText("Current Balance: ₹ " + balance));

    } catch (Exception e) {
        e.printStackTrace();
    }
    }).start();
}

```

```

private void loadTransactions() {

```

```

    new Thread(() -> {
        try {

            URL url = new URL(BASE_URL + "api/transactions");
            HttpURLConnection conn = (HttpURLConnection) url.openConnection();

            if (NetworkUtils.SESSION_COOKIE != null) {
                conn.setRequestProperty("Cookie",
                                        NetworkUtils.SESSION_COOKIE);
            }

            BufferedReader br =
                new BufferedReader(new InputStreamReader(conn.getInputStream()));

            StringBuilder response = new StringBuilder();
            String line;
            while ((line = br.readLine()) != null) {
                response.append(line);
            }
            br.close();

            JSONArray array =
                new JSONArray(response.toString());

```

```

runOnUiThread(() -> {
    adapter.clear();
    for (int i = 0; i < array.length(); i++) {
        try {
            JSONObject obj = array.getJSONObject(i);
            adapter.add(
                obj.getString("type") +
                " - ₹" +
                obj.getDouble("amount") +
                " - " +
                obj.getString("date"));
        } catch (JSONException e) {
            e.printStackTrace();
        }
    }
});

} catch (Exception e) {
    e.printStackTrace();
}
}).start();
}
}

```

NetworkUtils.java:

```

package com.example.bankapp;

public class NetworkUtils {
    public static String SESSION_COOKIE = null;
}

```